ERD Design – MoMo SMS Data Processing

The five entities are directly related to the concerns found in MoMo XML data. The central fact table is TRANSACTIONS, every XML record is a transaction with an amount, date, status and narrative description. USERS captures both sender and receiver in one table (not two tables), thus avoiding schema duplication; the sender/receiver distinction is carried by 2 FK's back to the TRANSACTIONS table. Because MoMo payments can be from many categories (incoming/outgoing transfers, merchant payments, bill payments, airtime top-ups), there is the need to separate TRANSACTION_CATEGORIES from the transaction itself so that these can be classified consistently and extended in the future. This is critical for audit trails in financial systems in order to track the ETL pipeline when an XML batch was ingested, if it was successful or if there was a parsing error.SYSTEM_LOGS is the tracking of the ETL pipeline, including the time an XML batch was ingested, whether it was a success or failure, and any parsing errors that occurred.
Multi Tagging: Each transaction may contain more than one analytical tag (e.g. "utility payment" and "recurring") and each category can be associated with many transactions. A meaningful entity because it has its own metadata (tagged_at, tagged_by) for this M:N relationship — not just a linking table.
Cardinalities: One sender and one receiver (mandatory FK), one category for its main type, and one log entry for each of its transactions. The ||--o{ notation illustrates that a

user can send/receive zero to many transactions, a category can apply to many transactions, and a log entry can apply to a batch of many transactions.
Data types are selected to conform to the precision of a monetary value (DECIMAL); to prevent losing leading zeros (VARCHAR for phone numbers); and for auditability (TIMESTAMP everywhere).